

Read the Fine Print: Assessing the User Experience of Reading in High Resolution Virtual Reality

Daniel Koch
daniel.koch@helsinki.fi
University of Helsinki
Helsinki, Finland

Alan Medlar
alan.j.medlar@helsinki.fi
University of Helsinki
Helsinki, Finland

Dorota Glowacka
dorota.glowacka@helsinki.fi
University of Helsinki
Helsinki, Finland

ABSTRACT

Despite the increasing visual fidelity of commercially available virtual reality (VR) headsets, they remain ill-suited to scenarios that require reading large quantities of text. Industrial VR headsets, however, have high resolution displays for immersive simulations, such as medical training, that could be repurposed to evaluate the potential future of reading in VR. In this article, we present a within-subject study on reading preferences and user experience comparing the high resolution Varjo VR2 Pro with the lower resolution Meta Quest 2. Our results show that choice of headset can have a significant impact on reading preferences, with an effect size comparable to interindividual differences. These results demonstrate that existing VR text guidelines are inappropriate at higher resolutions. Overall, however, users preferred the reading experience of the Meta Quest 2, citing the distracting border between high and lower resolution displays and the weight of the Varjo headset as impediments to reading.

CCS CONCEPTS

• Human-centered computing → User studies; Virtual reality.

KEYWORDS

Virtual reality, reading, user experience

ACM Reference Format:

Daniel Koch, Alan Medlar, and Dorota Glowacka. 2024. Read the Fine Print: Assessing the User Experience of Reading in High Resolution Virtual Reality. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3613905.3650916>

1 INTRODUCTION

Reading is a fundamental part of our everyday lives, facilitating education, knowledge work and leisure. Although reading is comfortable and well integrated into many digital devices, it has yet to become commonplace in virtual reality (VR). While VR remains ill-suited to reading large quantities of text, a newer generation of high resolution headsets intended for industrial applications have

the potential to support truly immersive reading (e.g., Varjo headsets¹). The majority of recent VR research, however, has focused on readily available consumer-grade headsets and it remains unclear whether previously published design guidelines for displaying text in VR [3, 12] are appropriate for higher resolutions.

In this article, we present a study on reading preferences and user experience in high resolution VR. We conducted a within-subject study comparing two headsets in the context of designing an adaptive e-Reader for VR: the Varjo VR2 Pro and the Meta Quest 2. The Varjo VR2 Pro has two screens per-eye: a high resolution focal display and a lower resolution peripheral display, whereas the Meta Quest 2 lacks the additional focal displays but is otherwise similar to the VR2 Pro. We wanted to contrast the optimal reading experience in each headset and to understand whether existing guidelines for displaying text in VR are valid for high resolution displays. Aside from display technologies, the two headsets have numerous differences, including whether they use hand-tracking or controllers for interaction. Therefore, we also measured users' perceived workload in finding the optimal reading experience and identified which physical headset characteristics impacted user experience through semi-structured interviews.

In brief, our results show that previously published guidelines for displaying text are headset-dependent, with the optimal font size decreasing as display resolution increases. Indeed, we show that choice of headset can have a significant impact on reading preferences, with an effect size comparable to interindividual differences. Last, we enumerate various aspects of the Varjo VR2 Pro that users felt degraded reading experience compared to the Meta Quest 2.

2 RELATED WORK

There is prior work on displaying text in VR related to textual attributes [3, 11–13], the positioning [2, 15, 17–19] and shape of the panel containing the text [20], how users interact with text [4, 14, 21] and related issues, such as learning outcomes [1].

The main studies related to our work assessed reading preferences and user experience in VR. Dingler et al. conducted the first study into reading preferences in VR and made design recommendations related to numerous attributes, including font size and text panel dimensions [3]. Kojić et al. conducted a similar study comparing two headsets, the Oculus Go and Oculus Quest, but found no significant differences in users' reading preferences [12]. Further studies have investigated reading preferences for Japanese [11] and Chinese [13] in VR. These studies are often self-described as guidelines for displaying text in VR, but whether their recommendations generalise beyond their own experimental setup remains unclear.

¹<https://varjo.com/>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3650916>

Aside from textual attributes, the relative position of the text panel, i.e., whether it has a world-fixed or head-fixed position, can impact both reading preferences and task performance. While early studies showed that head-fixed text panels improved task performance [2, 18], more recent studies have found that world-fixed text panels are preferred by users [11, 17]. Moreover, different head-fixed positions have been shown to impact perceived cognitive demand [15] and the design of panels in world-fixed positions can affect user experience [6]. More recently, Wei et al. investigated text panel curvature but found that users preferred flat panels over curved, cylindrical or spherical panels [20].

Other issues related to VR reading include: interaction methods, learning outcomes and display resolution. There have been numerous studies on how to interact with text in VR, including page turning [4], gaze-based magnification [14] and using VR as an assistive technology for visual disabilities [21]. Baceviciute et al. studied learning outcomes and found that reading in VR had better outcomes than listening to audio but was worse than reading physical books [1]. Lastly, Kilpeläinen and Häkkinen showed that text in the Varjo VR2 Pro is still legible at smaller font sizes than the HTC Vive and Meta Quest 2 [10]. To the best of our knowledge, this is the only VR reading study to investigate high resolution VR.

3 METHODOLOGY

We wanted to investigate reading preferences and user experience with two VR headsets: the Meta Quest 2 and the Varjo VR2 Pro. The Meta Quest 2 headset is currently the most popular consumer VR headset available, whereas the Varjo VR2 Pro is aimed at industrial applications and has more features, including high resolution displays and built-in eye-tracking. We designed a controlled laboratory study to compare both headsets in the context of designing an adaptive e-Reader for VR. Our research questions were as follows:

- RQ1: How does the display technology impact the optimal settings for reading experience?
- RQ2: How does the choice of headset impact perceived workload?
- RQ3: What physical attributes of each headset impact reading experience?

3.1 Participants

Twenty participants (5 female, 11 male, 3 non-binary, 1 prefer not to say) were recruited for our study. Participants were recruited using posters, email lists and via word of mouth. The median age of participants was 26, ranging from 18 to 33. A majority of participants had prior experience with VR (19/20), including games (16/20), movies (9/20), stereoscopic images (8/20) and other VR experiments (7/20). A quarter of participants owned a VR headset (5/20), but VR use was generally low (only 2/20 reported using VR either very often or everyday). Fewer than half of participants wore glasses or contact lens (7/20). We additionally asked participants about their reading habits. Half of the participants reported reading for pleasure either very often or everyday (10/20) and a minority favored reading physical over digital books (5/20).

3.2 Study Design

We designed a within-subject study to compare reading preferences in both VR headsets. We developed a virtual e-Reader interface that allowed participants to control various attributes of how text was displayed. Participants sat on a chair for the duration of the experiment to ensure they remained stationary and to avoid fatigue.

3.2.1 VR Headsets. The experiment was designed around the Varjo VR2 Pro headset and later ported to the Meta Quest 2. We break down the differences between headsets into two categories: display technologies and other physical characteristics:

- **Display technologies:** The Meta Quest 2 has an LCD display with a per eye resolution of 1832x1920 (90-120Hz). The Varjo VR2 Pro, however, has two displays per eye: a peripheral LCD display running at 1440x1600 (90Hz) and a focal display based on micro OLED running at 1920x1080 (60Hz)². The Meta Quest 2 has a greater visible field of view (97° horizontal FoV) compared to the Varjo VR2 Pro (87° horizontal FoV).
- **Physical characteristics:** The Varjo VR2 Pro is tethered to a host PC, whereas the Meta Quest 2 is stand-alone. The Varjo headset has a hard padded retractable strap and the Meta headset a elasticated fabric strap. Lastly, the Varjo headset is heavier than the Meta headset (605g and 503g, respectively).

3.2.2 Virtual e-Reader. Figure 1 shows the virtual e-Reader composed of a text panel controlled using virtual buttons that was implemented in Unity. The background color of the text panel was light gray and had 20px of padding. Participants were allowed to change the dimensions of the text panel (height and width), the viewing distance, and the font size. Participants were able to control each attribute using headset-specific gestures. To change the font size, for example, with the Varjo VR2 Pro participants would press the appropriate virtual button and then use two “okay” gestures to increase and decrease the font size. To perform the same action with the Meta Quest 2, the gestures were replaced with pressing both trigger buttons on the touch controllers (see Figure 2). We provided sufficient text to fill the panel, irrespective of the dimensions set by the user (an e-pub version of Jane Austin’s *Pride and Prejudice*). All text was displayed in Liberation Sans, an open source sans serif typeface metrically compatible with Helvetica and Arial.

Text rendering needed to be handled differently on each headset. Text on the Varjo VR2 Pro was rendered in-scene, but text on the Meta Quest 2 needed to be rendered on a separate compositor layer to reduce aliasing at higher viewing distances³.

3.3 Task

Participants were given a task to elicit their reading preferences with respect to the e-Reader’s attributes. Similar to [3], we set the following attributes: height, width, viewing distance and font size, to randomly sampled values and disabled one of the four attribute buttons. The initial attribute ranges were determined experimentally (height and width: 0.5-2.5m, distance: 0.8-7.0m, font size: 10-260pt). Participants were instructed to change the remaining three

²While the resolution of the focal displays are not significantly higher than the peripheral displays in terms of pixels, they are considerably smaller and, therefore, have higher pixel density. In this sense, we use the phrase “high resolution” to mean “high fidelity” throughout this article.

³<https://developer.oculus.com/resources/os-compositor-layers/>



Figure 1: A participant wearing a Varjo VR2 Pro headset and using the virtual e-Reader.

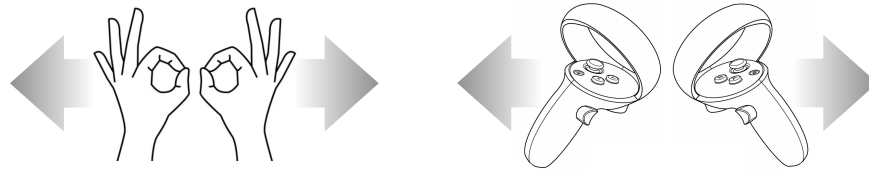


Figure 2: Gestures used to adjust the attributes of the virtual e-Reader. The Varjo headset used an “okay” gesture while moving both hands apart (together) to increase (decrease) an attribute (only increase shown). The Meta headset used the two trigger buttons on the touch controllers instead of a gesture.

attributes such that the text was comfortable to read without excessive horizontal or vertical head movements. Participants were provided with additional virtual buttons to indicate that they were satisfied with the current configuration or were unable to complete the task. Participants performed the same task several times using both headsets (see Section 3.5 below).

3.4 Measures

We collected the following measures: the e-Reader attributes set by participants during each task, responses from the NASA TLX questionnaire after using each headset [9], and semi-structured interviews conducted at the end of the study. The e-Reader attributes were measured in the units used by the Unity engine: height, width and viewing distance were measured in meters, and font size was measured in points. NASA TLX is a measure of cognitive workload that consists of 6 subscales for mental demand, physical demand, temporal demand, performance, effort and frustration. Each subscale consists of a single item on a 100-point scale in 5 point increments. We calculated the overall task load index assuming unweighted subscales (i.e., Raw NASA TLX) [8].

3.5 Procedure

We conducted a within-subject study to elicit participants’ reading preferences with the virtual e-Reader with both headsets. The order of headsets was balanced. At the time the study was conducted there were no longer any restrictions nor experimental guidelines related

to the COVID-19 pandemic at our university. Each VR headset was wiped down prior to the start of each experiment.

After providing informed consent, participants completed a demographics questionnaire and watched an instructional video where the task was explained to them. The video explained how to manipulate the attributes of the virtual e-Reader and the gestures required by each headset were demonstrated.

After watching the instructional video, the first VR headset was fitted to the participant’s head to ensure it was comfortable. Prior to performing the tasks, participants were encouraged to familiarize themselves with the hand-tracking/touch controllers, pressing virtual buttons and performing gestures to change the attributes of the e-Reader. When participants indicated they were ready, they were moved on to the experiment. Each participant performed 12 reading preference tasks with disabled attributes cycling through height, width, viewing distance and font size. For example, in the very first task, the height button was disabled and participants were only able to manipulate width, viewing distance and font size. The attribute order was not randomized because we wanted to maximize the time between tasks where a given attribute was disabled. After all tasks were completed, participants completed the NASA TLX questionnaire. The experiment was then repeated using the second VR headset.

After both experiments were completed, each participant was asked for feedback about their reading experiences with both headsets in a semi-structured interview. The complete study lasted approximately 60 minutes and participants received a book voucher for taking part in the study.

Reference	Headset	Preferred Size (dmm)	Preferred Size (pt@1m)
Dingler et al. [3]	Oculus Rift CV1	32 ± 11	132
Kojić et al. [12]	Oculus Go	16.01 ± 9.25	66
Kojić et al. [12]	Oculus Quest	17.87 ± 24.52	74
This article	Meta Quest 2	10.98 ± 6.90	45
This article	Varjo VR2 Pro	8.88 ± 4.71	37

Table 1: Summary of average preferred font sizes across publications and VR headset models. Average preferred font size is shown in distance-independent millimeters (dmm) as $mean \pm sd$ and the corresponding closest integer point size at a viewing distance of 1m.

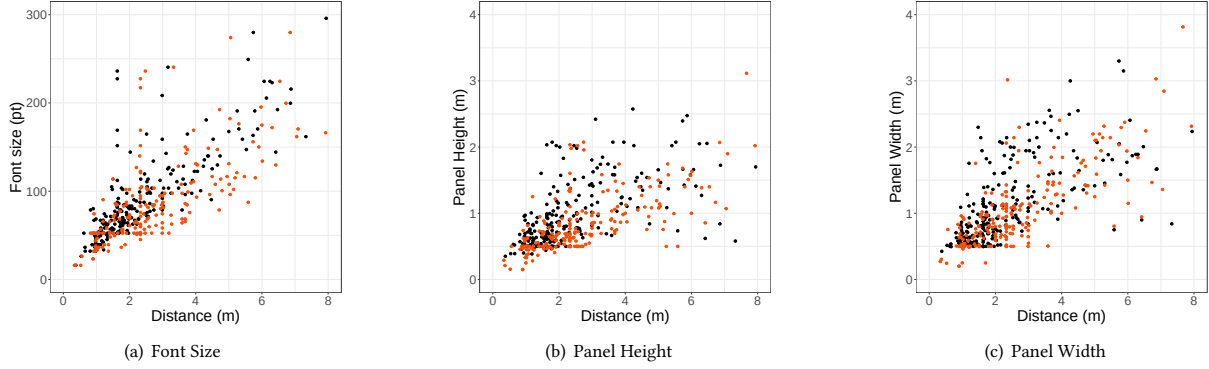


Figure 3: Raw preference data from Varjo VR2 Pro (red points) and Meta Quest 2 (black points). Height, width and font size were lower for Varjo VR2 Pro than Meta Quest 2 on average for the same viewing distance.

4 RESULTS

We analysed the reading preference data, NASA TLX responses and semi-structured interviews. We collected 448 observations of users' reading preferences over 19 participants. We needed to exclude P17 due to the log file not being saved for the Varjo headset and a further 8 observations were excluded due to participants indicating they were unable to complete the task. We did not record the reasons for each task failure, but they appeared to be due to extreme attribute values rather than physical discomfort, such as eyestrain, as participants continued the experiment without any further issues. The e-Reader attributes were all highly correlated with one another, with correlations ranging from 0.582-0.810 (Pearson's r^2). We, therefore, used a multivariate regression model to analyze the data (see below for further explanation) and η^2 to measure the effect size of each independent variable. As a majority of NASA TLX subscales were statistically significant according to the Shapiro-Wilk normality test, we used the non-parametric Wilcoxon signed-rank test to compare the cognitive workload for each headset and the point-biserial rank correlation coefficient to measure effect size.

4.1 Comparison with Previous Guidelines

Table 1 compares previously published guidelines for the angular size of text in distance-independent millimeters, where 1dmm is 1mm at a viewing distance of 1m. Angular size combines the font size (more precisely, the "cap height" of a given typeface) and the

viewing distance. We also report the equivalent font size for Liberation Sans at a viewing distance of 1m. We found that as the resolution of the headset increases, the optimal font size decreases, suggesting that guidelines for displaying text in VR need to be reevaluated for different display technologies. Anecdotally, we measured how close one of the authors held a physical book set in 10pt font away from their face while reading as ~ 8.0 dmm (very similar to the 8.88dmm measured with the Varjo headset), suggesting that for high resolution headsets we could start to use similar guidelines for VR as we do for print media.

4.2 Reading Preferences

We modeled reading preferences with a multivariate regression model to capture the joint variability between dependent variables. This reduces the standard errors on model coefficients compared to separate regression models, which increases statistical power. We had three dependent variables related to the virtual e-Reader (height, width and font size) and three independent variables (viewing distance, headset and participant ID). Height, width, viewing distance and font size were log-transformed due to being positive-only and to reduce the spread of observations at greater viewing distances (see Figure 3 for plots of raw data). The headset was modeled as a factor indicating whether the Varjo VR2 Pro or the Meta Quest 2 was used for a given observation.

Variable	Test stat. (Pillai trace, V)	P-value	Effect size (partial η^2)	Confidence interval (95%)
Viewing Distance	0.798	< 2.2e-16	0.80	[0.77, 0.82]
Participant	0.660	< 2.2e-16	0.22	[0.07, 0.23]
Headset	0.171	< 2.2e-16	0.17	[0.11, 0.23]

Table 2: MANOVA tests for variable inclusion, effect sizes and their 95% confidence intervals for reading preference model.

	Meta Quest 2		Varjo VR2 Pro		Test stat. (V)	P-value	Effect size (r)	Confidence interval (95%)
	Mean	SD	Mean	SD				
Mental demand	21.50	17.18	30.50	20.25	128.0	0.015*	-0.67	[-0.87, -0.28]
Physical demand	20.50	17.69	25.50	20.32	113.0	0.020*	-0.66	[-0.88, -0.22]
Temporal demand	15.75	15.41	19.00	19.51	55.0	0.218	-0.41	[-0.77, 0.14]
Performance	17.75	17.58	25.25	17.88	111.0	0.027*	-0.63	[-0.86, -0.19]
Effort	27.00	21.91	41.25	20.25	111.0	0.004**	-0.85	[-0.94, -0.62]
Frustration	20.75	17.27	27.50	22.45	103.5	0.069	-0.52	[-0.81, -0.02]
Overall	20.54	14.31	28.17	15.41	176.0	0.008**	-0.68	[-0.87, -0.29]

Table 3: Descriptive and test statistics for NASA TLX (* = statistically significant).

Table 2 shows the results from MANOVA tests for independent variable inclusion and their effect sizes. Based on these results, we can conclude that as well as viewing distance, the choice of headset and interindividual differences make a significant contribution to model fit, which was generally high with R^2 values of 0.638, 0.679 and 0.743 for the regression models for height, width and font size, respectively. Furthermore, the effect sizes for headset and participant are both “large” (following the interpretation guidelines for η^2 [5]), and of similar magnitude, pointing to the importance of personalization irrespective of headset⁴. Lastly, the sign of the headset coefficient was negative in each regression model (data not shown). This means that participants set the e-Reader’s height, width and font size lower for the Varjo VR2 Pro than the Meta Quest 2 after we control for viewing distance and participant. This is also visible in the raw data in Figure 3, where observations from the Varjo headset (red points), are frequently lower for the same viewing distance than the Meta headset (black points). This is due to the boundary between focal and peripheral displays on the Varjo headset which is more noticeable in high contrast settings, such as dark text on a light background.

4.3 Perceived Workload

Given the importance of personalization shown by users’ reading preferences, we wanted to understand how making adjustments to the virtual e-Reader differs between headsets in terms of perceived workload. Table 3 shows descriptive statistics and significance tests for the overall task load index and for each subscale of NASA TLX. The perceived workload was 37% higher in the Varjo VR2 Pro compared to the Meta Quest 2 ($V=176.0$, $P=0.008$). This was driven by significant increases in mental demand (37%), physical demand

(24%), performance (42%) and effort (53%) (see Table 3 for results of significance tests). We note, however, that the absolute score for the Varjo headset (28.17) was considerably lower than the median NASA TLX scores reported for many other tasks, including playing video games and driving a car [7].

4.4 Semi-structured Interviews

In the semi-structured interviews, we asked participants what they thought of each headset with respect to their reading experiences. We broke down their comments into three categories: visual quality, physical comfort and controls. Some participants referred to the Meta Quest 2 as the “Oculus Quest 2” as the headset has been sold under both names.

- **Visual quality:** Some participants said the Varjo headset had clearer text (4/20, e.g., “*Varjo sharp centre is better than Oculus for reading*”, P20), but others disliked it because of the visible boundaries between the focal and peripheral displays (5/20, e.g., “*Varjo text in the middle is better, but edges were annoying*” P6). Fewer participants stated that the Meta headset was clearer (2/20, e.g., “[Meta Quest 2 had a] *more clear picture*” P13). Only a single participant mentioned the aliasing issues⁵ on the Meta headset (“*Quest ... has more aliasing*” P19), however, other participants may have also noticed it, but lacked the vocabulary to describe what they were seeing.
- **Physical comfort:** Similar numbers of participants stated that the Varjo (3/20) and Meta headsets (2/10) were more comfortable. However, far more participants commented that they preferred the Meta headset because it weighed less (8/10, e.g., “*Oculus: lighter ... Varjo: too heavy*” P10).

⁴The effect size (η^2) is a standardized and, therefore, directly comparable with one another.

⁵See Section 3.2.2 for details.

- **Controls:** The most common issue highlighted by participants was about hand-tracking vs physical controls. Half of the participants stated that they preferred the physical controls of the Meta headset's touch controllers (10/20, e.g., "*hand-tracking is okay, but [touch] controllers more accurate and smoother*" P19). A single participant saw the benefit of using hand-tracking while reading (1/20, "[it was a] *plus when no physical controllers*" P1).

5 DISCUSSION AND FUTURE WORK

In this article, we presented a study of reading preferences and user experience in VR using the high resolution Varjo VR2 Pro and the lower resolution Meta Quest 2.

In RQ1, we investigated how different display technologies impact reading preferences. We showed that previously published guidelines for displaying text in VR are headset-dependent and that for high resolution headsets like the Varjo VR2 Pro participants' preferred a font size similar to that used in print media. Compared to the Meta Quest 2, however, participants preferred shorter and narrower text panels with the Varjo headset. This difference highlights the limitations of separate focal and peripheral displays for reading in VR, with many participants (5/20) describing the boundary between displays as "annoying". We note that this boundary is particularly noticeable in high contrast scenarios, such as dark text on light backgrounds, which could make it generally inappropriate for reading in VR for longer periods of time. Last, we showed that the choice of headset can have a significant impact on reading preferences, with an effect size comparable to participants' interindividual differences. This suggests that taking into account the specific features and limitations of a VR headset when designing a reading application is as important for user experience as incorporating personalization features.

In RQ2 and RQ3, we were interested in how the choice of headset impacted perceived workload and which physical attributes of each headset affected reading experience. Achieving the optimal reading experience with the Varjo headset had a perceived workload 37% higher than the Meta Quest 2. This difference in perceived workload was strongest for mental demand, which was likely caused by a combination of unfamiliar controls (i.e., using hand-tracking rather than physical controllers) and ensuring the panel dimensions did not interfere with the boundary between focal and peripheral displays. We note, however, that while the Varjo headset had a higher perceived workload than the Meta Quest 2, it was not high compared to other everyday tasks [7]. In interviews participants gave varied responses to the question of which physical attributes impacted their experiences. Fewer than a third of participants mentioned the visual fidelity of one headset over the other (6/20), with more participants preferring Varjo's (4/20) than Meta's (2/20) displays. Indeed, in our interviews the biggest differentiators between headsets were weight (8/20) and controls (10/20) with both being worse for the Varjo headset, which corresponds to its greater physical demand score in NASA TLX. These findings highlight the importance of the physical characteristics of VR headsets, especially as the fidelity of display technologies approaches the limits discernible by the human eye.

The study we conducted had several limitations. First, the age range of participants was relatively young and was, therefore, not

representative of the general population. Second, we decided to use the default control method for each headset (hand-tracking with Varjo, controllers with Meta), however, the issues participants raised about controls might have overshadowed other equally valid critiques that were otherwise missed. Third, as we wanted to collect as many observations from each participant as possible, they did not spend a lot of time with each e-Reader configuration, which may have obscured the influence of longer term issues, such as eye strain, that could have impacted our results. In future work, we want to incorporate the models developed in this article with information-rich virtual environments. Recent work has shown that subtle reorientation of the camera during teleportation can increase user engagement with points of interest [16] and we want to investigate whether similar techniques could be used to draw attention to text in VR.

ACKNOWLEDGMENTS

This work has been supported by Helsinki Institute for Information Technology (HIIT).

REFERENCES

- [1] Sarune Baceviciute, Aske Mottelson, Thomas Terkildsen, and Guido Makransky. 2020. Investigating representation of text and audio in educational VR using learning outcomes and EEG. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376872>
- [2] Jian Chen, P.S. Pyla, and D.A. Bowman. 2004. Testbed evaluation of navigation and text display techniques in an information-rich virtual environment. In *IEEE Virtual Reality 2004*. 181–289. <https://doi.org/10.1109/VR.2004.1310072>
- [3] Tilman Dingler, Kai Kunze, and Benjamin Outram. 2018. VR reading UIs: Assessing text parameters for reading in VR. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems (CHI EA '18)*. Association for Computing Machinery, New York, NY, USA, 1–6. <https://doi.org/10.1145/3170427.3188695>
- [4] Tilman Dingler, Siran Li, Niels van Berkel, and Vassilis Kostakos. 2021. Page-turning techniques for reading interfaces in virtual environments. In *Proceedings of the 32nd Australian Conference on Human-Computer Interaction (OzCHI '20)*. Association for Computing Machinery, New York, NY, USA, 454–461. <https://doi.org/10.1145/3441000.3441024>
- [5] Andy Field. 2013. *Discovering statistics using IBM SPSS statistics*. Sage.
- [6] Jenny Gabel, Melanie Ludwig, and Frank Steinicke. 2023. Immersive reading: Comparison of performance and user experience for reading long texts in virtual reality. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*. Association for Computing Machinery, New York, NY, USA, 1–8. <https://doi.org/10.1145/3544549.3585895>
- [7] Rebecca A. Grier. 2015. How high is high? A meta-analysis of NASA-TLX Global workload scores. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* 59, 1, 1727–1731. <https://doi.org/10.1177/1541931215591373>
- [8] Sandra G. Hart. 2006. NASA-Task Load Index (NASA-TLX); 20 Years Later. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* 50, 9, 904–908. <https://doi.org/10.1177/154193120605000909>
- [9] Sandra G. Hart and Lowell E. Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research. In *Human Mental Workload*. Advances in Psychology, Vol. 52. North-Holland, 139–183. [https://doi.org/10.1016/S0166-4115\(08\)62386-9](https://doi.org/10.1016/S0166-4115(08)62386-9)
- [10] Markku Kilpeläinen and Jukka Häkkinen. 2023. An effective method for measuring text legibility in XR devices reveals clear differences between three devices. *Frontiers in Virtual Reality* 4 (2023). <https://doi.org/10.3389/frvir.2023.1243387>
- [11] Seina Kobayashi, Kei Kanari, and Mie Sato. 2021. An examination of view-settings for long texts in VR reading. In *ACM SIGGRAPH 2021 Posters*. 1–2.
- [12] Tanja Kojić, Danish Ali, Robert Greinacher, Sebastian Möller, and Jan-Niklas Voigt-Antons. 2020. User experience of reading in virtual reality – Finding values for text distance, size and contrast. In *2020 Twelfth International Conference on Quality of Multimedia Experience (QoMEX)*. 1–6. <https://doi.org/10.1109/QoMEX48832.2020.9123091>
- [13] Alex Pak Ki Kwok, Mian Yan, Zhe Hao Xu, Jing Yun Lin, Run Da Chen, and Shi Long Wen. 2023. Effects of line length, number of lines, line spacing, and font size on reading performance of Chinese text in virtual reality environment. *International Journal of Industrial Ergonomics* 96 (2023), 103469. <https://doi.org/10.1016/j.ergon.2023.103469>

- [14] Geonsun Lee, Jennifer Healey, and Dinesh Manocha. 2022. VRDoc: Gaze-based interactions for VR reading experience. In *2022 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 787–796. <https://doi.org/10.1109/ISMAR55827.2022.00097>
- [15] Nils Adrian Mack, Markus Heun, and Marion Rose. 2023. Head-anchored text placements and cognitive load in information-rich virtual environments. In *Proceedings of Mensch Und Computer 2023*. Association for Computing Machinery, New York, NY, USA, 27–36. <https://doi.org/10.1145/3603555.3603575>
- [16] Alan Medlar, Mari Lehtikari, and Dorota Glowacka. 2024. Behind the scenes: Adapting cinematography and editing concepts to navigation in virtual reality. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24)*. Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3613904.3642412>
- [17] Jason Orlosky, Kiyoshi Kiyokawa, and Haruo Takemura. 2014. Managing mobile text in head mounted displays: Studies on visual preference and text placement. *SIGMOBILE Mob. Comput. Commun. Rev.* 18, 2 (jun 2014), 20–31. <https://doi.org/10.1145/2636242.2636246>
- [18] Nicholas F. Polys, Seonho Kim, and Doug A. Bowman. 2005. Effects of information layout, screen size, and field of view on user performance in information-rich virtual environments. In *Proceedings of the ACM Symposium on Virtual Reality Software and Technology* (Monterey, CA, USA) (VRST '05). Association for Computing Machinery, New York, NY, USA, 46–55. <https://doi.org/10.1145/1101616.1101626>
- [19] Rufat Rzayev, Polina Ugnivenko, Sarah Graf, Valentin Schwind, and Niels Henze. 2021. Reading in VR: The effect of text presentation type and location. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3411764.3445606>
- [20] Chunxue Wei, Difeng Yu, and Tilman Dingler. 2020. Reading on 3D surfaces in virtual environments. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. 721–728. <https://doi.org/10.1109/VR46266.2020.00095>
- [21] Kurtis Weir, Fernando Loizides, Vinita Nahar, Amar Aggoun, and Andrew Pollard. 2023. I see therefore I read: Improving the reading capabilities of individuals with visual disabilities through immersive virtual reality. *Universal Access in the Information Society* 22, 2 (2023), 387–413. <https://doi.org/10.1007/s10209-021-00854-8>